SMALL MODULI PROBLEMS OVER $\operatorname{Spec}\mathbb{Z}$: QUADRATIC ALGEBRAS, ROTATIONS IN PGL_2 AND THE INTEGRAL SCHWARZIAN, AFTER GROTHENDIECK'S WORKING NOTES

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Abstract. In the early 1980s Grothendieck worked out, in notes now kept at the University of Montpellier, several small moduli problems over an arbitrary base, without inverting 2, 3 or 5. We state and prove what can be proved of them in elementary terms, completed or corrected where the pages are incomplete.

Quadratic algebras $R[U]/(U^2 + bU + c)$ are classified, when 2 is a non-zero-divisor, by triples (L, δ, T_0) with $T_0^2 \equiv \delta \bmod 4$. Their product, generated by $2u_1u_2 + b_1u_2 + b_2u_1$, has discriminant $\delta_1\delta_2$, and it is the module of coinvariants of $\sigma_1 \otimes \sigma_2$ exactly when 2, b_1, b_2 generate the unit ideal.

The Hamilton–Cayley algebra structures on $H \otimes H'$ extending two quadratic algebras form an affine line. A structure is Azumaya exactly when $Q(t) = t^2 - bb't + b^2c' + b'^2c - 4cc'$ is a unit, over every ring, characteristic 2 included, and the degenerate locus receives the product of the two quadratic algebras.

For a rotation g in a form of PGL_2, the invariant $A(g) = (\operatorname{Tr} g)^2/\det g - 2$ detects the orders 3 to 6 over any base, with g^5 scalar exactly when $A(g)^2 + A(g) = 1$. Rigidity and a normal form are established for the string group G_2. In characteristic 5 the octahedral configuration of $\mathbb{P}^1$ is all of $\mathbb{P}^1(\mathbb{F}_5)$, which breaks the identification of its group with the octahedral group.

The Schwarzian written with divided derivatives, $G_0 = G/12$, is a cocycle over any base. Its vanishing forces a homography at Frobenius level p for $p > 3$, but only at level p^2 for $p = 2, 3$.

All statements proved here are formalised in Lean; what remains of Grothendieck's moduli statements is listed.

1. Introduction

Five folders of Grothendieck's working notes, kept at the University of Montpellier, treat small moduli problems over an arbitrary base:

- folder 84, on binary quadratic forms and quadratic algebras, dated from 1982 to about 1986 by the inventory;
- folder 83, on Azumaya algebras of rank 4, with a part dated April 1986;
- folder 85, *2-polyèdres et 1-polygones*, on regular polygons and polyhedra realised in forms of PGL_2;
- folder 64, *Modules courbes elliptiques en caractéristiques* $\neq 2$, dated from 1980–1982;
- folder 93, on the Schwarzian derivative and projective connections.

They share a method: write the objects in coordinates over $\mathbb{Z}$, find the invariants, and see what survives at the primes 2, 3 and 5 that the classical theory inverts. This paper states and proves the results of these folders that can be established in elementary terms. The statements are attributed to Grothendieck when the page states them, with the page; otherwise they are said to be *after* the page, and the change is described. The proofs are ours unless we say that we follow the page. Every result proved here is formalised in Lean 4 with Mathlib (Section 6).

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Grothendieck's global moduli statements, which need more than coordinates, are quoted as statements, with what we prove of them.

Throughout, R is a commutative ring.

2. Quadratic algebras and their product

Folder 84 works with quadratic algebras $A(b,c) = R[U]/(U^2 + bU + c)$, without dividing by 2, and their discriminant $\delta(b,c) = b^2 - 4c$. On a general base, a quadratic cover is given by an invertible module L (the quotient $A/\mathcal{O}$, up to sign), $b \in L$ and $c \in L \otimes L$; its *triple* is (L, δ, T_0), where $\delta = b \otimes b - 4c$ and T_0 is the class of b modulo $2L$. A morphism of covers is a pair (φ, t) with $\varphi\colon L \to L'$, $t \in L'$, $\varphi(b) = b' + 2t$ and $(\varphi \otimes \varphi)(c) = c' + b' \otimes t + t \otimes t$.

Theorem 2.1 (Grothendieck, folder 84, pp. 45–53, affine base)**.**

1. *Over every R, the triple of a cover satisfies $T_0^2 \equiv \delta \bmod 4$, independently of the lift of T_0, and morphisms of covers induce morphisms of triples.*
2. *If 2 is a non-zero-divisor in R, every morphism of triples lifts to a unique morphism of covers, and every triple (L, δ, T_0) with $T_0^2 \equiv \delta \bmod 4$ comes from a cover. The functor from covers to triples is an equivalence of groupoids, compatible with base change.*
3. *Over $\mathbb{Z}$, an integer δ is a discriminant iff $\delta \equiv 0, 1 \bmod 4$, and $A(b,c) \cong A(b',c')$ iff $\delta(b,c) = \delta(b',c')$. Over $\mathbb{F}_2$, $A(1,0) = \mathbb{F}_2[U]/(U^2+U)$ and $A(1,1)$ have the same b and the same δ and are not isomorphic, so (2) fails without the hypothesis.*

Sketch. $b^2 - \delta = 4c$, and $(\tau + 2a)^2 - \delta = \tau^2 - \delta + 4(a\tau + a^2)$. When 2 is regular, the equation $\varphi(b) = b' + 2t$ determines t, and the second condition follows from $\varphi(\delta) = \delta'$ after division by 4, which is legitimate since 4 is regular. Pages 49–52 reconstruct the algebra, its trace and its norm from (L, δ, T_0) without division: an element y has coordinates (β, γ) with $\beta \equiv \gamma b \bmod 2$, the unit is $(2, 0)$, and $2\beta(yy') = \beta\beta' + \gamma\gamma'\delta$, $2\gamma(yy') = \beta\gamma' + \gamma\beta'$, $4N(y) = \beta^2 - \gamma^2\delta$. □

The product. Page 63 defines the product of two quadratic algebras "sauf erreur" by $b = b_1 b_2$ and $4c = b^2 - \delta_1\delta_2$; the boxed formula writes $4c_1c_1$ where the computation gives $4c_1c_2$. Page 64 checks the generator.

Proposition 2.2 (Grothendieck, folder 84, pp. 63, 64, 89, 102)**.** *Let u_i satisfy $u_i^2 + b_i u_i + c_i = 0$ in commutative R-algebras A_i, and put $b = b_1 b_2$, $c = c_1 b_2^2 + c_2 b_1^2 - 4c_1c_2$. In any commutative ring, with $Q_i = u_i^2 + b_i u_i + c_i$ and $u = 2u_1u_2 + b_1u_2 + b_2u_1$,*

$$u^2 + bu + c = b_2^2 Q_1 + b_1^2 Q_2 + 4(Q_1Q_2 - c_1Q_2 - c_2Q_1).$$

Hence $\mathcal{U} = 2U_1U_2 + b_1U_2 + b_2U_1 \in A_1 \otimes_R A_2$ satisfies $\mathcal{U}^2 + b\mathcal{U} + c = 0$, and defines a morphism $A(b,c) \to A_1 \otimes A_2$. Moreover $b^2 - 4c = \delta_1\delta_2$.

Proof. Expand both sides; the identity is polynomial in u_1, u_2, b_i, c_i. □

The product of quadratic algebras over a general base, with multiplicative discriminant, is studied by Voight as the monoid of quadratic rings [5]; see also Knus [4] for quadratic algebras over rings. Grothendieck's pages identify the product with the invariants of the involution $\sigma_1 \otimes \sigma_2$, where σ_i is the canonical involution of A_i.

Theorem 2.3 (Grothendieck, folder 84, pp. 84–86)**.** *For split A_i with bases $(1, u_i)$, and E_i the dual modules:*

1. *The induced map $(E_1 \otimes E_2)/\operatorname{Im}(\sigma_1 \otimes \sigma_2 - \mathrm{id}) \to E_1 * E_2$ is well defined, and it is bijective iff $2R + b_1R + b_2R = R$.*
2. *The map $A(b,c) \to (A_1 \otimes A_2)^{\sigma_1 \otimes \sigma_2}$ is an isomorphism after every base change $R \to S$ iff $2R + b_1R + b_2R = R$. Over $R = \mathbb{Z}[X]$ with $b_1 = X$, $b_2 = 0$, it is an isomorphism over R itself, although $(2, X, 0)$ is a proper ideal: the condition "universally" is needed.*

In geometric terms: the product is the invariant algebra, universally, iff at every point of residue characteristic 2 one of the two algebras is étale. Pages 65–79 build the product of extensions $0 \to L \to E \to M \to 0$ equipped with a χ-splitting, for a fixed $\chi \in R$, by an amalgamated sum, without division.

Theorem 2.4 (Grothendieck, folder 84, pp. 65–79 and 123–131, split case)**.** *For any R and $\chi \in R$, zero included, split extensions $M \oplus L$ with the χ-splitting of a map $b\colon M \to L$, with morphisms (f, φ, w) satisfying $\varphi b = b' f + \chi w$, form a symmetric monoidal category under $b_1 \otimes b_2$; a morphism is invertible as soon as f and φ are. For $M = R$, the object (R, L, b) is invertible iff L is invertible and $L = Rb + \chi L$. The "pushed" and "pulled" products of page 125 are identified by a transvection $T_{\gamma b}$ when $\gamma\chi = 2$, and for the unit objects an identification exists iff such a γ exists; over $\mathbb{Z}$ with $\chi = 4$ there is none.*

3. The line of twists and its degenerate locus

Folder 83 constructs, for $(b, c, b', c', t) \in R^5$, the algebra $M = M(b, c; b', c'; t)$, free with basis $1, u, v, uv$, with $u^2 = bu - c$, $v^2 = b'v - c'$ and $\mathrm{Tr}(uv) = t$; pages 126–136 take $H = R[u]/(u^2 - bu + c)$, $H' = R[v]/(v^2 - b'v + c')$ and study the algebra structures on $H \otimes H'$. Put

$$Q(t) = t^2 - bb't + b^2c' + b'^2c - 4cc'.$$

Theorem 3.1 (after folder 83, pp. 81–84, 92–98 and 130–136)**.**

1. *For every t, M is an associative unital R-algebra inducing H and H', every element satisfies $x^2 - \mathrm{Tr}(x)x + \det(x) = 0$, $\det$ is multiplicative, and $\mathrm{Tr}(uv) = t$. Conversely, in any R-algebra whose elements satisfy such a Hamilton–Cayley equation, two elements u, v with the traces and norms above span the image of a multiplicative map from some $M(b, c; b', c'; t)$, with t unique if the map is injective. Distinct t give distinct structures: the structures form the affine line of t.*
2. (Grothendieck, pp. 97–98.) *Over every commutative ring, characteristic 2 included and without hypothesis on δ, δ', M is an Azumaya algebra iff $Q(t)$ is a unit. Equivalently, the form $\det$ on $B = R1 + Ru + Rv$ has invertible half-discriminant, which equals $-Q(t)$.*
3. *The discriminant of the trace form of M in the basis $(1, u, v, uv)$ is $-Q(t)^2$. The polynomial Q has discriminant $\delta\delta'$ and is invariant under $t \mapsto bb' - t$, and the degenerate locus $\widetilde{S} = \mathrm{Spec}\, R[t]/(Q)$ is free of rank 2 over R.*
4. *In $H \otimes H'$, the element $u * v = b'u + bv - 2u \otimes v$ satisfies $Q(u * v) = 0$, which gives a morphism $R[t]/(Q) \to H \otimes H'$; the canonical involutions send $u * v$ to $bb' - u * v$. For characters φ, φ' of H, H' with values in a commutative R-algebra K, $(\varphi \otimes \varphi')(u * v) = b'\varphi(u) + b\varphi'(v) - 2\varphi(u)\varphi'(v)$ is a root of Q.*

Point (4) is the link with Section 2: up to the sign conventions, $u * v$ is the generator $\mathcal{U}$ of the product of H and H', and Q is its equation. So the non-Azumaya locus of the line of twists is the product of the two quadratic algebras, as pages 137–143 assert, and the map $X \times_S X' \to \widetilde{S}$ of the page is the canonical morphism of Proposition 2.2. For $b = c = b' = c' = 0$, $M(0, 0; 0, 0; t)$ is Azumaya iff t is a unit (page 97).

4. Rotations in forms of PGL_2

Folder 85 realises polygons and polyhedra in a form of PGL_2 over a base by a single invariant. For $g \in M_2(R)$ with $\det g$ invertible, put

$$A(g) = \frac{(\mathrm{Tr}\, g)^2}{\det g} - 2,$$

which depends only on the image of g in $\mathrm{PGL}_2(R)$, and is $\zeta + \zeta^{-1}$ when g has eigenvalue ratio ζ. Call g *nowhere scalar* if its image in $M_2(R/\mathfrak{m})$ is not scalar for every maximal ideal $\mathfrak{m}$.

Theorem 4.1 (after folder 85, pp. 6–7, 10, 29–31 and 40)**.** *Let $g \in M_2(R)$ be nowhere scalar, with trace t, invertible determinant d, and $\alpha = A(g)$.*

1. *With $x_0 = 0$, $x_1 = 1$, $x_{n+2} = tx_{n+1} - dx_n$, one has $g^{n+1} = x_{n+1}g - dx_n$, and g^{n+1} is scalar iff $x_{n+1} = 0$.*
2. *g^3 is scalar iff $\alpha = -1$; g has order 4 at every point iff $\alpha = 0$ and 2 is invertible; g has order 6 at every point iff $\alpha = 1$ and 6 is invertible.*

(3) g^5 is scalar iff $\alpha^2+\alpha-1=0$. Moreover $A(g^2)=A(g)^2-2$, so that $A(g^2)=-1-A(g)$ in that case (the corollary of page 40).

Proof. (1) is Cayley–Hamilton, $g^2=tg-d$, and g nowhere scalar. Writing $x_{2k+1}=d^kP_{2k+1}(\alpha)$ and $x_{2k+2}=t\,d^kP_{2k+2}(\alpha)$ with $P_3=\alpha+1$, $P_4=\alpha$, $P_5=\alpha^2+\alpha-1$, $P_6=\alpha^2-1$ gives (2) and (3); the even cases need t invertible, which is where 2 and 6 enter. $A(g^2)=(t^2-2d)^2/d^2-2=A(g)^2-2$. □

Example 4.2 (folder 85, p. 40). Over $\mathbb{Z}[\zeta_5]$, the matrices $s=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$ and $p=\left(\begin{smallmatrix}\zeta&1\\0&\zeta^4\end{smallmatrix}\right)$ satisfy $s^2=-1$, $(sp)^3=-1$, p^5 scalar, and $A(p)=\zeta^2+\zeta^3$, a root of T^2+T-1; s, p and sp are nowhere scalar. They give an icosahedral subgroup of $\mathrm{PGL}_2(\mathbb{Z}[\zeta_5])$.

Page 40 states:

Statement 4.3 (Grothendieck, folder 85, p. 40)**.** For a form G of PGL_2 over a scheme S and a monomorphism $\varphi\colon\mathfrak{A}_5\to G$, the element $\alpha(\varphi)=A(\varphi(\pi))$, π a 5-cycle, satisfies $\alpha^2+\alpha-1=0$; the category of pairs (G,φ) is rigid, and its functor of isomorphism classes is represented by $\operatorname{Spec}\mathbb{Z}[T]/(T^2+T-1)$.

Theorem 4.1(3) proves the first assertion over every base, at the level of matrices. Rigidity and representability are not proved here; they would give icosahedral subgroups unique up to unique isomorphism over residue characteristics 2, 3 and 5, once a root of T^2+T-1 is chosen. For finite subgroups of $\mathrm{PGL}_2(K)$ over fields, see Beauville [2].

The string group G_2. Pages 4–7 treat homomorphisms of $G_2=\langle\tau_0,\tau_1,\tau_2\mid\tau_i^2,(\tau_0\tau_2)^2\rangle$ into $\mathrm{PGL}_2(R)$, given by matrices τ_i with invertible determinants and τ_i^2, $(\tau_0\tau_2)^2$ scalar. Put $\rho_0=\tau_0\tau_1$, $\rho_1=\tau_1\tau_2$, $\alpha_i=A(\rho_i)$, and $\Delta=(\alpha_0+2)(\alpha_1+2)(\alpha_0+\alpha_1)$. Over a field, condition $(*)$ of page 4 says that ρ_0^2 and ρ_1^2 are not scalar and that $\rho_0\rho_1$ is not a scalar multiple of $\rho_1\rho_0$.

Theorem 4.4 (after folder 85, pp. 4–7)**.**

(1) $(*)$ holds at every point iff τ_0, τ_1, τ_2 and $\tau_0\tau_2$ are traceless and Δ is invertible; each factor of Δ matches one condition.

(2) Rigidity. *Under these conditions, if $g\in\mathrm{GL}_2(R)$ and $\mu_i\in R$ satisfy $g\tau_i=\mu_i\tau_i g$ for $i=0,1,2$, then g is scalar.*

(3) Normal form. *For every (α_0,α_1) with Δ invertible, explicit traceless matrices with these invariants exist over $R[T]/(T^2-\alpha_0T+1)$, free of rank 2 over R; over a field containing a root of $T^2-\alpha_0T+1$, two triples with the same invariants are conjugate in PGL_2.*

The resulting moduli statement, that these homomorphisms into forms of PGL_2 are classified by the complement of the three lines $\Delta=0$ in $\mathbb{A}^2_{\mathbb{Z}}$, is the edition's reformulation of pages 4–7 and is not proved here.

Octahedral configurations. Folder 64 considers a faithful action of the Klein group on $\mathbb{P}^1$, locally $z\mapsto-z$, $z\mapsto1/z$, whose axes form the octahedral configuration $\Delta=\{0,\infty,\pm1,\pm i\}$. To identify $\mathrm{Aut}(\mathbb{P}^1,\Delta)$ with the rotation group of the octahedron, page 48 computes the fixer of the non-opposite vertices ∞ and -1, whose elements are $z\mapsto az+(a-1)$.

Theorem 4.5 (Grothendieck, folder 64, p. 48)**.** *Let K be a field with $2\neq0$ and $i^2=-1$. If $z\mapsto az+(a-1)$ maps $\{0,1,i,-i\}$ into itself, then $5(a-1)=0$. In particular, if $5\neq0$ in K, the fixer of ∞ and -1 in $\mathrm{Aut}(\mathbb{P}^1_K,\Delta)$ is trivial.*

Proof. The map is an automorphism, so $a\neq0$ and it is injective; mapping the four-element set $\{0,1,i,-i\}$ into itself, it permutes it. The four points sum to 1, and their images sum to $a+4(a-1)=5a-4$, so $5a-4=1$. □

The margin restricts the computation to "car. $\neq3$ et 5"; characteristic 3 needs no special case. In characteristic 5 the identification fails: $z\mapsto2z+1$ permutes $\{0,1,i,-i\}$, and over $\mathbb{F}_5$, where $i=2$, the configuration Δ is the whole of $\mathbb{P}^1(\mathbb{F}_5)$, so all 120 elements of $\mathrm{PGL}_2(\mathbb{F}_5)$

preserve it, and the fixer of ∞ and -1 has four elements. Pages 44–52 state that, over a base with residue characteristics $\neq 2$, subgroup schemes of $\mathrm{Aut}_S(X)$ locally isomorphic to $(\mathbb{Z}/2)^2$, those locally isomorphic to $\mathfrak{S}_4$, and regular octahedral configurations correspond bijectively; Theorem 4.5 shows that the hypothesis "5 invertible" is needed. Pages 74–91 state a moduli description of elliptic curves with a level-2 structure and a "Legendre pinning" over $\mathbb{Z}[1/2]$, by $U_{0,3} \times \mu_4^*$; for the Brauer group and related structures of the moduli stack of elliptic curves, see Antieau and Meier [1]. Neither statement is proved here.

5. The Schwarzian with divided derivatives

Folder 93 studies the Schwarzian derivative and projective connections on a smooth curve over a base. For two uniformisers t, s, the classical Schwarzian difference is $G(t,s) = \big(-2s'''/s' + 3(s''/s')^2\big)\,dt^2$. Page 72 replaces derivatives by the divided operators $D^{(i)} = \frac{1}{i!}\frac{d^i}{dt^i}$ of Hasse and Schmidt [3], observes that the coefficients 2 and 3 forbid them, and takes $G_0 = G/12$:

$$G_0(t,s) = -\frac{D^{(3)}s}{D^{(1)}s} + \Big(\frac{D^{(2)}s}{D^{(1)}s}\Big)^2.$$

Put $N(s) = (D^{(2)}s)^2 - D^{(1)}s \cdot D^{(3)}s$, so that $G_0(s) = N(s)/(D^{(1)}s)^2$.

Theorem 5.1 (Grothendieck, folder 93, pp. 71–73 and 115–123)**.**

(1) $G = 12\,G_0$*, and* G_0 *is defined over every base, characteristics* 2 *and* 3 *included.*

(2) Cocycle. *For polynomials* s*,* u*:* $N(s\circ u) = (N(s)\circ u)(D^{(1)}u)^4 + \big((D^{(1)}s)\circ u\big)^2 N(u)$*; if* $D^{(1)}s$ *and* $D^{(1)}u$ *are invertible,* $G_0(s\circ u) = (G_0(s)\circ u)(D^{(1)}u)^2 + G_0(u)$*.*

(3) Homographies. *If* $a,b,c,d \in R$ *and* $s(cX+d) = aX+b$ *with* $cX+d$ *invertible, then* $G_0(s) = 0$*.*

(4) Killed functions. *In characteristic* p*, a polynomial is killed by* $D^{(1)}$*,* $D^{(2)}$*,* $D^{(3)}$ *iff it is a polynomial in* X^p *for* $p>3$*, in* X^4 *for* $p=2$*, in* X^9 *for* $p=3$*.*

(5) Rigidity. *Let* A *be a local ring of characteristic* p*,* $N=1$ *if* $p>3$ *and* $N=2$ *if* $p=2,3$*, and* $\alpha \in A$*. An* $s \in A[T]/(T^{p^N}-\alpha)$ *with* $D^{(1)}s$ *invertible and* $G_0(t,s)=0$ *is* $(at+b)/(ct+d)$ *with* $a,b,c,d \in A$ *and* $ct+d$ *invertible.*

Sketch. (1)–(3) are identities in the divided-power calculus. For (5) with $\alpha = 0$, the coefficient of X^j in $q\,N(s)$, $q = cX+d$, is $-\binom{j+3}{3}s_1 r_{j+3}$ when X^{j+3} divides $r-(aX+b)$, $r = sq$; with $c = -s_2/s_1$, $b = s_0$, $a = s_1 + cs_0$, induction on j gives $X^n \mid s(cX+1)-(aX+b)$ as long as $\binom{k}{3}$ is invertible for $3 \le k < n$, which holds for $n = p$ ($p>3$), $n=4$ ($p=2$) and $n=9$ ($p=3$). For general α over a local ring, base-change to $B = A[T]/(T^{p^N}-\alpha)$, where $T^{p^N}-\alpha = (T-\beta)^{p^N}$, and use the Taylor map $f \mapsto f(\beta+Y)$. □

Two corrections to the pages. The determinant cannot be normalised to $ad-bc=1$, as page 121 writes: in $\mathbb{F}_5[T]/(T^5)$, $s = 2t$ has $N(s) = 0$ and $D^{(1)}s = 2$ invertible, but 2 is not a square in $\mathbb{F}_5$, and s is no homography with $ad-bc=1$. And the page passes from $\alpha \neq 0$ to $\alpha = 0$ without argument ("OPS"); the base change above supplies it over a local ring, where $\mathrm{Pic}(A) = 0$. The jump from level p to level p^2 at $p = 2,3$ is the shadow of the factor 12: G_0 sees the third divided derivative, which vanishes on X^p only when $p > 3$.

6. Verification

The statements proved above are formalised in Lean 4 with Mathlib in the files `Folder84.lean`, `Folder83.lean`, `Folder85.lean`, `Folder64.lean` and `Folder93.lean` of the repository Commutator-IO/grothendieck-project (directory `lean/Grothendieck`), using only Lean's standard axioms; the symmetric monoidal structure of Theorem 2.4 uses Mathlib's `MonoidalCategory` and `SymmetricCategory`. Not proved here: the global form of Theorem 2.1 on schemes; the intrinsic product of non-split χ-extensions; rigidity and representability in Statement 4.3 and in the G_2 moduli; the relative octahedral correspondence and the Legendre pinnings of folder 64.

Statement on the use of artificial intelligence

The transcriptions of folders 64, 83, 84, 85 and 93, the modernised readings, the proofs, the Lean formalisation and a first draft of this text were produced with language models (Claude Opus 5 and Opus 5.5, Anthropic), under the author's direction. The author checked the statements, the proofs and the references, and takes full responsibility for the content. The formal proofs are checked mechanically by Lean.

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References

[1] B. Antieau, L. Meier, The Brauer group of the moduli stack of elliptic curves, *Algebra Number Theory* 14 (2020), 2295–2333, doi:10.2140/ant.2020.14.2295.

[2] A. Beauville, Finite subgroups of $\mathrm{PGL}_2(K)$, in *Vector Bundles and Complex Geometry*, Contemp. Math. 522, Amer. Math. Soc., 2010, 23–29, doi:10.1090/conm/522/10289.

[3] H. Hasse, F. K. Schmidt, Noch eine Begründung der Theorie der höheren Differentialquotienten in einem algebraischen Funktionenkörper einer Unbestimmten, *J. reine angew. Math.* 177 (1937), 215–237, doi:10.1515/crll.1937.177.215.

[4] M.-A. Knus, *Quadratic and Hermitian Forms over Rings*, Grundlehren der mathematischen Wissenschaften 294, Springer, 1991, doi:10.1007/978-3-642-75401-2.

[5] J. Voight, Discriminants and the monoid of quadratic rings, *Pacific J. Math.* 283 (2016), 483–510, doi:10.2140/pjm.2016.283.483.

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